Abstract

Contemporary web tracking continues to rapidly evolve as browsers and organizations move away from traditional third-party cookies. While web tracking has heavily relied on a stateful cookie-based approach, stateless tracking methods and other innovative technologies provide organizations with new ways to monitor and obtain user information.

This study will conduct a cross-genre quantitative analysis of approximately 120 publicly accessible websites. An automated Python program using Selenium and Google Chrome will collect a standard set of measurements related to stateful and stateless tracking, third-party tracking, and cookie security attributes, including Secure, HttpOnly, and SameSite. A sample of the automated measurements will be manually verified to evaluate the accuracy of the collection process. Pandas will be used to organize and analyze the collected data and compare tracking and security characteristics across the website genres. Selected measurements will also be incorporated into a simple predictive classification model to evaluate patterns in website tracking behavior. In addition, major U.S. privacy laws and regulatory frameworks will be examined to find how their protections apply to emerging web-tracking technologies.

This research seeks to answer three primary questions: (1) what technologies are surfacing as alternatives to third-party cookies? (2) How broadly do websites implement recommended cookie security attributes and practices, including Secure, HttpOnly, and SameSite? (3) to what extent have the major U.S. privacy laws and regulatory frameworks kept pace with the demand to protect users' data.